



Software Testing And Quality Assurance

Group3 - RegNo:

26919/2024

26497/2024

26886/2024

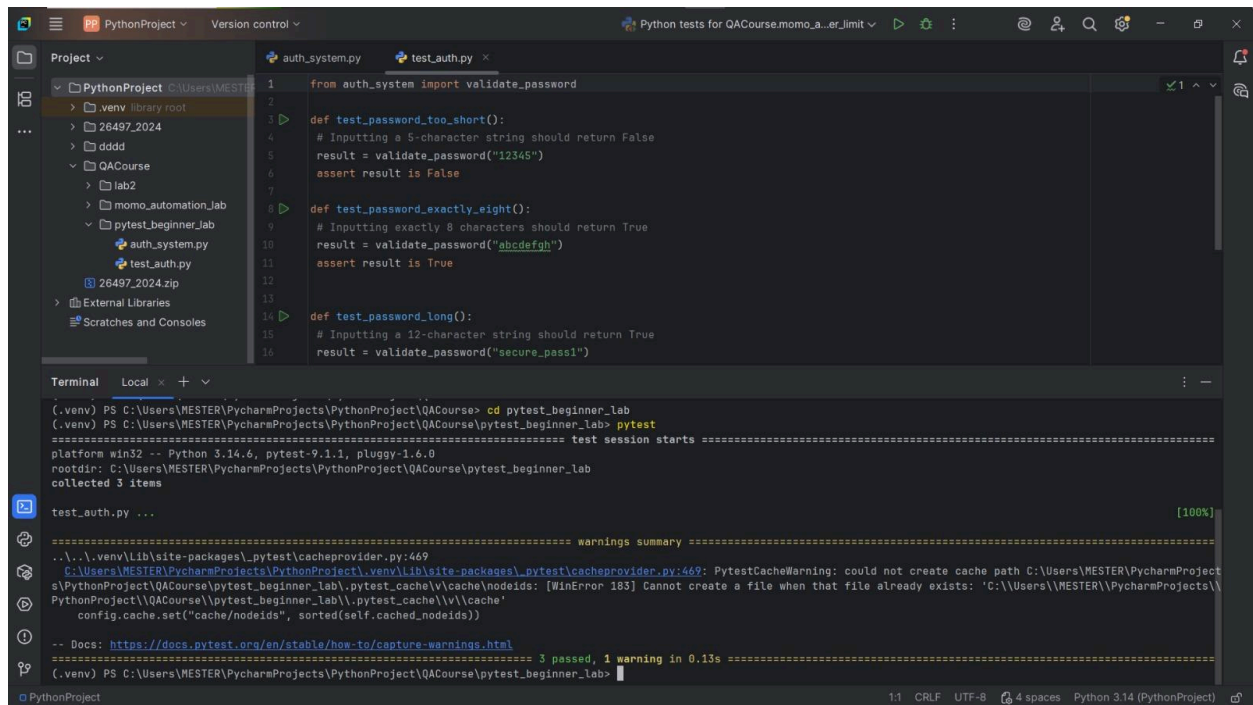
25952/2024

24761/2024

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4. Step 3: Implement the Verification Module



The screenshot shows the PyCharm IDE with the `test_auth.py` file open. The code defines three test functions: `test_password_too_short()`, `test_password_exactly_eight()`, and `test_password_long()`. The terminal output shows the test session starting and passing all three tests.

```
1 from auth_system import validate_password
2
3 def test_password_too_short():
4     # Inputting a 5-character string should return False
5     result = validate_password("12345")
6     assert result is False
7
8 def test_password_exactly_eight():
9     # Inputting exactly 8 characters should return True
10    result = validate_password("abcdefgh")
11    assert result is True
12
13
14 def test_password_long():
15    # Inputting a 12-character string should return True
16    result = validate_password("secure_pass1")
```

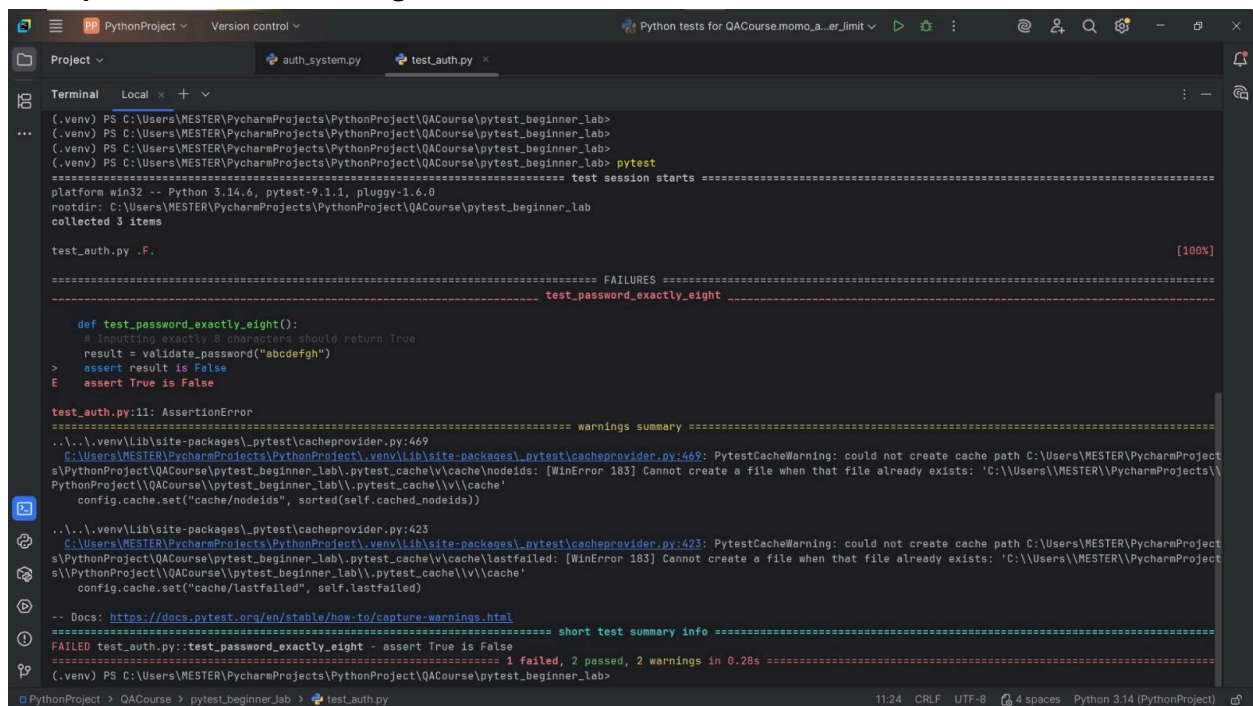
```
(.venv) PS C:\Users\MESTER\PycharmProjects\PythonProject\QACourse> cd pytest_beginner_lab
(.venv) PS C:\Users\MESTER\PycharmProjects\PythonProject\QACourse\pytest_beginner_lab> pytest
===== test session starts =====
platform win32 -- Python 3.14.6, pytest-9.1.1, pluggy-1.6.0
rootdir: C:\Users\MESTER\PycharmProjects\PythonProject\QACourse\pytest_beginner_lab
collected 3 items

test_auth.py ... [100%]

===== warnings summary =====
..\..\venv\Lib\site-packages\_pytest\cacheprovider.py:469
  C:\Users\MESTER\PycharmProjects\PythonProject\venv\Lib\site-packages\_pytest\cacheprovider.py:469: PytestCacheWarning: could not create cache path C:\Users\MESTER\PycharmProject
s\PythonProject\QACourse\pytest_beginner_lab\pytest_cache\cache\nodeids: [WinError 183] Cannot create a file when that file already exists: 'C:\Users\MESTER\PycharmProjects\
PythonProject\QACourse\pytest_beginner_lab\pytest_cache\cache'
    config.cache.set("cache/nodeids", sorted(self.cached_nodeids))

-- Docs: https://docs.pytest.org/en/stable/how-to/capture-warnings.html
===== 3 passed, 1 warning in 0.13s =====
(.venv) PS C:\Users\MESTER\PycharmProjects\PythonProject\QACourse\pytest_beginner_lab>
```

5. Step 4: Run Suite & Debug Failures



The screenshot shows the PyCharm IDE with the `test_auth.py` file open. The terminal output shows the test session starting and failing one test: `test_password_exactly_eight`. The debug console shows the assertion error and the warning summary.

```
(.venv) PS C:\Users\MESTER\PycharmProjects\PythonProject\QACourse\pytest_beginner_lab>
(.venv) PS C:\Users\MESTER\PycharmProjects\PythonProject\QACourse\pytest_beginner_lab>
(.venv) PS C:\Users\MESTER\PycharmProjects\PythonProject\QACourse\pytest_beginner_lab>
(.venv) PS C:\Users\MESTER\PycharmProjects\PythonProject\QACourse\pytest_beginner_lab> pytest
===== test session starts =====
platform win32 -- Python 3.14.6, pytest-9.1.1, pluggy-1.6.0
rootdir: C:\Users\MESTER\PycharmProjects\PythonProject\QACourse\pytest_beginner_lab
collected 3 items

test_auth.py .F. [100%]

===== FAILURES =====
..... test_password_exactly_eight .....

def test_password_exactly_eight():
    # Inputting exactly 8 characters should return True
    result = validate_password("abcdefgh")
    > assert result is False
E   assert True is False

test_auth.py:11: AssertionError
===== warnings summary =====
..\..\venv\Lib\site-packages\_pytest\cacheprovider.py:469
  C:\Users\MESTER\PycharmProjects\PythonProject\venv\Lib\site-packages\_pytest\cacheprovider.py:469: PytestCacheWarning: could not create cache path C:\Users\MESTER\PycharmProject
s\PythonProject\QACourse\pytest_beginner_lab\pytest_cache\cache\nodeids: [WinError 183] Cannot create a file when that file already exists: 'C:\Users\MESTER\PycharmProject
s\PythonProject\QACourse\pytest_beginner_lab\pytest_cache\cache'
    config.cache.set("cache/nodeids", sorted(self.cached_nodeids))

..\..\venv\Lib\site-packages\_pytest\cacheprovider.py:423
  C:\Users\MESTER\PycharmProjects\PythonProject\venv\Lib\site-packages\_pytest\cacheprovider.py:423: PytestCacheWarning: could not create cache path C:\Users\MESTER\PycharmProject
s\PythonProject\QACourse\pytest_beginner_lab\pytest_cache\cache\lastfailed: [WinError 183] Cannot create a file when that file already exists: 'C:\Users\MESTER\PycharmProject
s\PythonProject\QACourse\pytest_beginner_lab\pytest_cache\cache'
    config.cache.set("cache/lastfailed", self.lastfailed)

-- Docs: https://docs.pytest.org/en/stable/how-to/capture-warnings.html
===== short test summary info =====
FAILED test_auth.py::test_password_exactly_eight - assert True is False
===== 1 failed, 2 passed, 2 warnings in 0.28s =====
(.venv) PS C:\Users\MESTER\PycharmProjects\PythonProject\QACourse\pytest_beginner_lab>
```

After changing `assert result is False` for function `test_password_exactly_eight()` After changing

6. Student Lab Challenge Assignment

Terminal Local x

```
(.venv) PS C:\Users\MESTER\PycharmProjects\PythonProject\QACourse\pytest_beginner_lab> pytest
===== test session starts =====
platform win32 -- Python 3.14.6, pytest-9.1.1, pluggy-1.6.0
rootdir: C:\Users\MESTER\PycharmProjects\PythonProject\QACourse\pytest_beginner_lab
collected 6 items

test_auth.py .....

[100%]

===== warnings summary =====
..\..\venv\Lib\site-packages\_pytest\cacheprovider.py:469
  C:\Users\MESTER\PycharmProjects\PythonProject\venv\Lib\site-packages\_pytest\cacheprovider.py:469: PytestCache
Warning: could not create cache path C:\Users\MESTER\PycharmProjects\PythonProject\QACourse\pytest_beginner_lab\
pytest_cache\v\cache\nodeids: [WinError 183] Cannot create a file when that file already exists: 'C:\Users\MEST
ER\PycharmProjects\PythonProject\QACourse\pytest_beginner_lab\pytest_cache\v\cache'
    config.cache.set("cache/nodeids", sorted(self.cached_nodeids))

-- Docs: https://docs.pytest.org/en/stable/how-to/capture-warnings.html
===== 6 passed, 1 warning in 0.07s =====
(.venv) PS C:\Users\MESTER\PycharmProjects\PythonProject\QACourse\pytest_beginner_lab> 
```

PythonProject > QACourse > pytest_beginner_lab > test_auth.py

28:1

CRLF

UTF-8

4 spaces

Python 3.14 (PythonProject)